

Chapter 9 — From fundamentals to advanced collection

Chapter 2 covered surveys, sampling, and small-scale collection

Learning objectives

- By the end of this part

Decision focus still anchors advanced pipelines

- Advanced collection is not an excuse to gather everything
- Teams still need the right evidence for a concrete action
- Crowdsourcing, sensors

Example 9.1 — Customer Sentiment Survey for Decisions

- Example 9.1 — hands-on module
- Example 9.1 considers a product team that may still begin with customer-satisfaction
- Once the same questions must be answered continuously across regions, channels
- Advanced methods preserve that decision-making goal at larger scale
- Explore the chapter example module
- View files: ``modules/chapter9/example1/``

Why traditional pipelines hit limits

- Traditional collection often fails on three fronts: scalability, reliability, and cost
- Paper and interview workflows grow expensive as samples expand
- Those limits motivate the techniques developed later in this chapter

Example 9.2 — Paper Surveys and Scalability Limits

- Example 9.2 — hands-on module
- Example 9.2 shows how paper surveys fail to scale with sample size
- Distribution, collection
- The lesson is operational: when turnaround and volume matter, manual logistics break first
- Explore the chapter example module
- View files: ``modules/chapter9/example2/``

Heterogeneous sources and continuous streams

- Beyond scale and cost
- These sources differ in structure, arrive at high velocity
- Quality assurance therefore depends on declared crawl windows, stable identifiers

Example 9.5 — Wearables as Continuous Health Data Sources

- Example 9.5 — hands-on module
- Example 9.5 shows wearables as continuous health sources that traditional clinic visits
- Fitness trackers and smartwatches stream heart rate, activity
- Explore the chapter example module
- View files: ``modules/chapter9/example5/``

Takeaways

- Advanced collection extends Chapter 2 when data grow too large
- Decision focus remains the design goal
- Crowdsourcing, IoT and edge sensing

Next

- Complete the quiz for this part
- The next part defines crowdsourcing and walks use cases from fraud-pattern review and